



---

**Experiment 7: A grayscale image is converted into a digital image.  
Write a Python program to perform image quantization.**

**Code:**

```
from PIL import Image
# Open grayscale image
img=Image.open(r"C:\Users\mohit\OneDrive\Pictures\MSD.jpg").convert(
"L")
# Number of gray levels
levels = 4
# Quantization step
step = 256 // levels
# Perform quantization
quantized = img.point(lambda p: (p // step) * step)
# Save output image
quantized.save("quantized_output.jpg")
print("Image quantization completed!")
print("Output saved as quantized_output.jpg")
```

**Input:**



**Output:**



---

**Experiment 1: A camera captures an image that appears blurred due to improper sampling. Write a Python program to demonstrate image sampling.**

**Code:**

```
import cv2
# Read the image
img = cv2.imread("image1.png")
if img is None:
    print("Image not found!")
else:
    # Downsample
    small = cv2.resize(img, (150, 150))
    # Upsample
    sampled = cv2.resize(small, (img.shape[1], img.shape[0]))
    # Display images
    cv2.imshow("Original Image", img)
    cv2.imshow("Sampled Image", sampled)
    # Save output
    cv2.imwrite("sampled_image.jpg", sampled)
    print("Sampled image saved successfully!")
    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

**Sample input:**



**Output:**

